Correlation Analysis between Agricultural CO2 Emissions and Average Temperature Changes

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1 Introduction

One of the most urgent issues facing the world today is climate change, which has an impact on many facets of life, such as biodiversity, sea levels, and weather patterns. An important factor contributing to climate change is the rise in greenhouse gases, especially carbon dioxide (CO2). A number of agricultural processes, including soil cultivation, livestock management, and deforestation, significantly increase CO2 emissions. For the purpose of creating sustainable agricultural practices and successful climate policies, it is essential to comprehend the connection between temperature changes and agricultural CO2 emissions. The objective of this report is to examine the relationship between changes in average temperature and agricultural CO2 emissions across a number of nations between 2000 and 2010.

2 Used Data

For this analysis, two comprehensive datasets were utilized:

2.1 Agri-food CO2 Emission Dataset

- **Source:** Kaggle
- **Description:** This dataset includes detailed information on CO2 emissions from various agricultural activities, such as savanna fires, forest fires, crop residues, and rice cultivation. The data is aggregated from the Food and Agriculture Organization (FAO) and the Intergovernmental Panel on Climate Change (IPCC) reports, providing a comprehensive overview of agricultural emissions.
- **Data Structure:** The dataset contains columns representing the year, country (referred to as 'Area'), and different sources of CO2 emissions, including total emissions.
- **Data Licensing:** The dataset is available under the terms specified by the data providers (FAO and IPCC) and should be used in compliance with their data usage policies.

2.2 Climate Change - Earth Surface Temperature Data

- **Source:** Kaggle
- **Description:** This dataset provides global land temperature data, aggregated by country, sourced from the Berkeley Earth Surface Temperature Study. It includes average temperatures, as well as maximum and minimum temperatures, and their uncertainties.
- **Data Structure:** The dataset contains columns for the date (referred to as 'dt'), average temperature, and country.
- **Data Licensing:** The dataset is available under the terms specified by the Berkeley Earth project and should be used in compliance with their data usage policies.

3 Analysis

The analysis process involved several key steps:

3.1 Data Collection and Cleaning

The datasets were downloaded using the Kaggle API, and relevant CSV files were read into dataframes using the Pandas library in Python. The CO2 emissions data was filtered to include only the years 2000 to 2010. Similarly, the temperature data was filtered to the same period. Both datasets were cleaned and processed to ensure consistency and accuracy, including handling missing values and ensuring correct data types.

3.2 Data Aggregation and Integration

The temperature data was aggregated by year and country to compute the average temperature. This involved converting the date column to datetime format and extracting the year. The CO2 emissions data was also aggregated by year and country. The two datasets were then merged on the common columns, 'Year' and 'Country', to create a unified dataframe for analysis.

3.3 Statistical Correlation Analysis

A statistical analysis was conducted to determine the correlation between CO2 emissions and average temperature. The Pearson correlation coefficient was calculated to quantify the strength of the relationship. The resulting correlation coefficient was found to be -0.06, indicating a weak negative correlation between CO2 emissions and average temperature changes. This suggests that during the period from 2000 to 2010, increased agricultural CO2 emissions were not strongly associated with changes in average temperature.

4 Conclusions

The analysis reveals a weak negative correlation between agricultural CO2 emissions and average temperature changes from 2000 to 2010. This suggests that during this period, increased agricultural CO2 emissions were not strongly associated with changes in average

temperature. However, it is essential to consider other factors and longer time frames for a comprehensive understanding of the relationship.

4.1 Critical Reflection

While the analysis provides insights into the relationship between agricultural CO2 emissions and temperature changes, there are several limitations to consider:

- **Temporal Scope:** The analysis is limited to a decade (2000-2010), which may not be sufficient to capture long-term trends.
- **Data Granularity:** The aggregation of data by year and country may overlook regional and seasonal variations.
- **Other Factors:** The analysis focuses solely on CO2 emissions and temperature, without considering other greenhouse gases, climatic factors, or socio-economic variables that could influence the relationship.

Future studies could include more variables and advanced modeling techniques to provide deeper insights into the relationship between agricultural activities and climate change. Additionally, extending the temporal scope and incorporating regional and seasonal analyses could offer a more nuanced understanding of the impact of agricultural emissions on temperature changes.